

Defense 4: In-Network Normalization of a DNP3 Response-Time Fingerprint

Corrected binary on silicon — accepted timing evidence

2026-08-07

Summary

On one Intel Tofino-1 switch between a DNP3 master and a physical SEL-751 relay, Defense 4 holds the relay’s acknowledgment and response in the switch and releases them on a schedule, without changing any byte. On the corrected switch program, verified from the loaded configuration (binary **97175e7d**), two independent campaigns of 600 transactions each show that the response-deadline and dual-deadline modes turn the relay’s wide native response-time fingerprint into a fixed 10 ms value, holding every response, with a small late tail reported honestly. Every result below is scored by a fail-closed pipeline (78 adversarial self-tests) and reproduces across fixed and randomized block order.

The observable and the mechanism

The cross-layer response time (CLRT) is the master-facing interval from the pure TCP acknowledgment to the first byte of the matching DNP3 response. A passive observer measures it over many polls as a device fingerprint.

Defense 4 keeps the original acknowledgment and response queue-resident while higher-priority internal blocker tokens recirculate, and the traffic manager releases them on schedule. Four strict-priority queues guarantee the acknowledgment is never released after the response.

Five modes are configurations of the one framework: OFF (native), D1 (event), D2 (response deadline, D_A=0), D3 (acknowledgment deadline, D_R=0), D4 (dual deadline). The next figure shows, from the measured medians, exactly where each mode releases the acknowledgment and the response, and the CLRT that results.

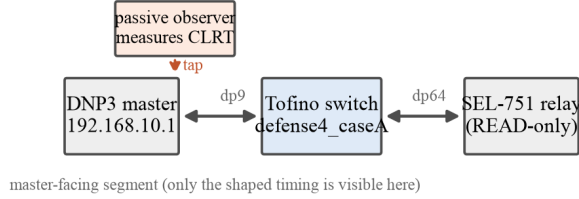


Figure 1: Testbed and adversary placement. The switch sits between the master and the relay; the passive observer taps the master-facing segment, where only the shaped timing is visible.

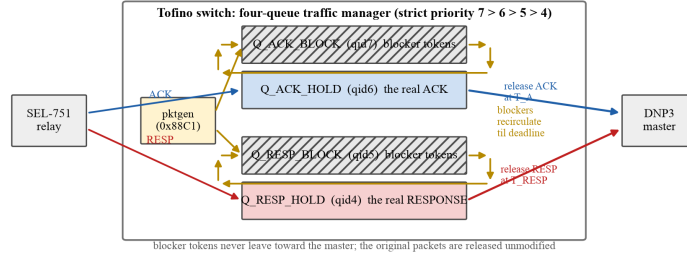


Figure 2: The four-queue hold-and-release mechanism. The relay's acknowledgment and response wait in low-priority hold queues (qid6, qid4); pktgen blocker tokens recirculate in the high-priority block queues (qid7, qid5) until the deadline; the traffic manager then releases the originals. Strict priority 7>6>5>4 orders the acknowledgment before the response, and the blocker tokens never leave toward the master.

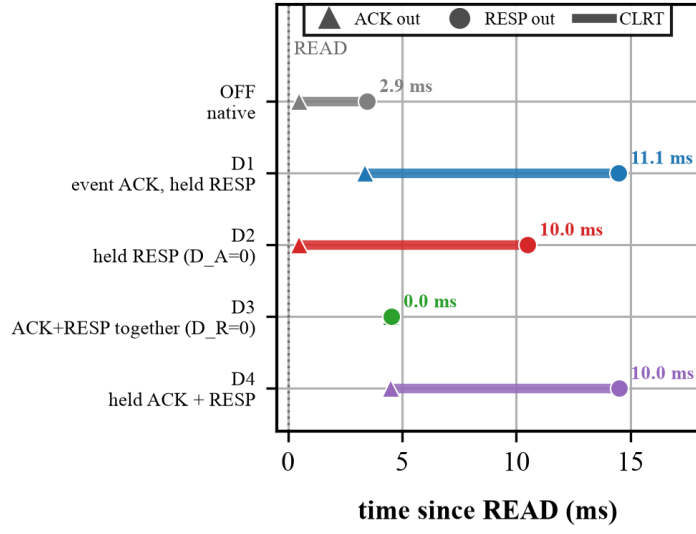


Figure 3: Per-mode timing sequence, from the measured median read-to-acknowledgment and read-to-response over both campaigns. OFF is native and narrow; D2 and D4 pin the response so the CLRT is a fixed 10 ms; D3 releases both together (CLRT ~ 0); D1 pins the response but releases the acknowledgment on an event, so its CLRT is wide.

Result: the CLRT is normalized, and the fingerprint information collapses

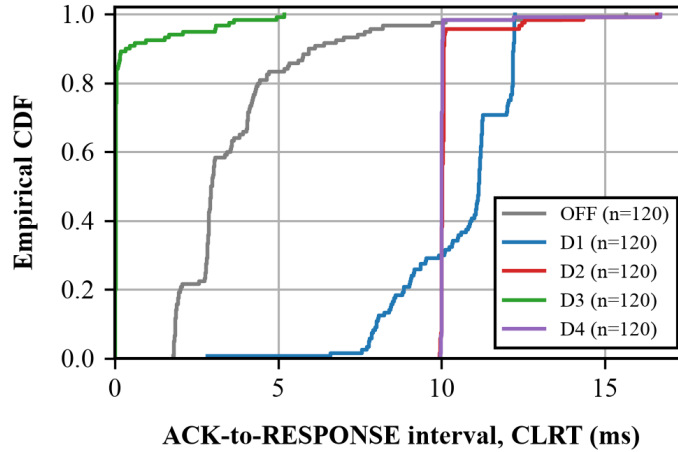


Figure 4: Native (OFF) versus the Defense 4 modes: empirical CDF of the CLRT. OFF is the wide native curve; D2 and D4 are near-vertical steps at 10 ms with a short late tail; D3 collapses to ~ 0 ; D1 shifts to ~ 11 ms. $n=120$ per mode, Campaign A.

Across both campaigns ($n = 240$ per mode) the must-hold modes held every response ($\text{RESP_BYPASS} = 0$), where the pre-fix binary bypassed D2 240/240 and D4 80/240. The CLRT dispersion and information collapse:

mode	p5-p95 spread (ms)	spread reduction vs OFF	CLRT entropy (bits)	effective timing states
OFF	5.69	1.0x	3.63	12.4
D2	0.12	45.6x	1.23	2.3
D4	0.05	118.2x	1.10	2.1
D3	1.05	5.4x	0.76	1.7
D1	5.17	1.1x	3.39	10.5

D4 cuts the CLRT spread 118 times and the observable’s entropy from about 12 effective timing states to about 2. D1 (event) is the weakest normalizer: it moves the timing without tightening it.

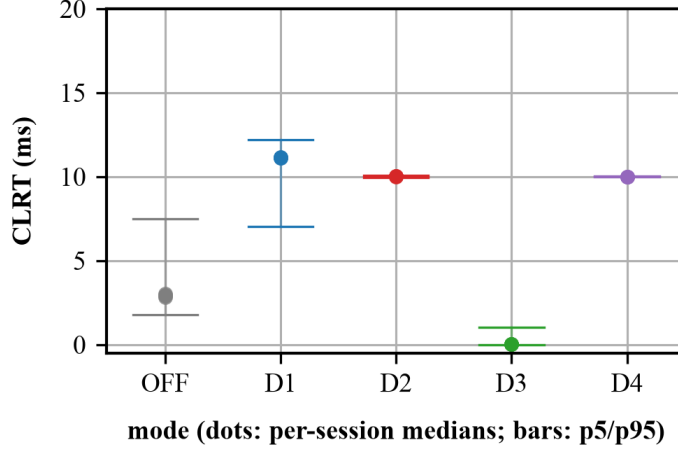


Figure 5: Per-session medians (dots) and pooled p5/p95 (bars) per mode. D2 and D4 are tight at 10 ms and stable across sessions; D1 (event) shifts the distribution but keeps its spread.

Honest boundary

We report the whole distribution: D2 and D4 normalize the bulk, not every observation (a small fraction arrive late and are released safely, D4 max about 16.7 ms). This is timing normalization for one separate-acknowledgment device, not a cross-device fingerprint-defeat claim, which needs a second comparable device. The controlled negative-case tests are built and unit-tested offline but not yet run live; the paired byte-identity comparator is validated on crafted captures; there is no final acceptance verdict, and the paper Introduction stays quarantined.